Rozilyn Marco

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ABOUT ME

Robotics researcher designing compliant robots that can sense physical contact. My MASc work embeds magnetic sensing in tendon-driven continuum robots to detect and localize external contact, combining sensor-integrated mechanism design with learning-based estimation, toward safer teleoperation and contact-aware control.

EDUCATION AND TRAINING

28/08/2025 – 27/08/2027

MASTER OF APPLIED SCIENCE (MASC) IN MECHANICAL ENGINEERING University of Toronto

Continuum Robotics Laboratory (Supervisor: Professor Jessica Burgner-Kahrs)

31/08/2019 – 14/06/2025

BACHELOR OF APPLIED SCIENCE (BASC) IN ENGINEERING SCIENCE University of Toronto

Final grade 3.51/4.00 (With Honours) | **Thesis** An Open-Source, Extensible, Tendon-Driven Continuum Robotic System

PUBLICATIONS

Automated Planning Domain Inference for Task and Motion Planning

Jinbang Huang, Allen Tao, **Rozilyn Marco**, Miroslav Bogdanovic, Jonathan Kelly, Florian Shkurti
International Conference on Robotics and Automation (ICRA), 2025

Robotic Food Handling Utilizing Temperature Dependent Variable-Stiffness Material

Rozilyn Marco, Prashant Kumar, Xinyi Zhang, Weiwei Wan, Kensuke Harada
International Conference on Artificial Life and Robotics (ICAROB), 2024

WORK EXPERIENCE

GRADUATE RESEARCHER WITH PROFESSOR JESSICA BURGNER-KAHRS – CONTINUUM ROBOTICS LABORATORY, UNIVERSITY OF TORONTO MISSISSAUGA – 31/07/2025 – Current – CANADA

- Conduct research on the design, modeling, and control of tendon-driven continuum robots.
- Develop mechanical prototypes using multi-material 3D printing and perform experimental characterization of curvature, force–displacement behaviour, and workspace metrics.
- Contribute to an open-source continuum robotics platform through validated fabrication processes and reproducible testing procedures.

RESEARCH INTERN WITH PROFESSOR JONATHAN KELLY – STARS LABORATORY, UNIVERSITY OF TORONTO – 31/03/2024 – 30/06/2024 – CANADA

- Contributed to a deep learning framework integrating Graph Attention Networks with automated planning for task-domain inference.
- Developed and validated an RGB-D pose estimation pipeline for extracting task conditions for PDDL generation.
- Evaluated learned planning domains on a Franka Research 3 robot for multi-step manipulation tasks.
- Co-authored a publication accepted to ICRA 2025 and delivered the spotlight presentation and poster session.

ROBOTIC RESEARCH INTERN WITH PROFESSOR KENSUKE HARADA – ROBOTIC MANIPULATION LAB, OSAKA UNIVERSITY – 30/03/2023 – 29/06/2023 – JAPAN

- Designed and prototyped a variable-stiffness end-effector using deformable materials.
- Conducted mechanical characterization experiments involving temperature-dependent stiffness changes and force-gauge measurements.
- Refined designs based on experimental results to improve force transmission and stability.
- Work led to a first-author publication at ICAROB 2024.

DATA ANALYST WITH PROFESSOR EMILY MOORE – INSTITUTE FOR STUDIES IN TRANSDISCIPLINARY ENGINEERING, UNIVERSITY OF TORONTO – 30/07/2022 – Current

- Designed and maintained engagement databases (2,400+ records), improving data quality and traceability across internal systems.
- Implemented automated data-processing pipelines in R and developed analytical dashboards (Excel, Power BI) to support program evaluation and evidence-based decision-making.

HONOURS AND AWARDS

22/07/2026

First Runner-Up, Poster Session Competition – Toronto Robotics Conference, University of Toronto Robotics Institute

- Awarded for research on contact sensing for continuum robots, presented in the conference's poster session competition.

Link https://crl.utm.utoronto.ca/2026/07/23/trc.html

06/07/2026

Ontario Graduate Scholarship (OGS) – Government of Ontario and University of Toronto

Merit-based provincial graduate scholarship ($15,000) awarded for outstanding academic achievement, supporting MASc research in continuum robotics.

01/05/2024

NSERC Undergraduate Student Research Award (USRA) – Natural Sciences and Engineering Research Council of Canada (NSERC)

- Competitive national award funding a full-time 16-week undergraduate research project.

01/09/2023

First Place - International IEEE Solid-State Circuits Society Arduino Contest – Institute of Electrical and Electronics Engineers (IEEE)

- Awarded for technical excellence and innovation in hardware–software system design.

01/05/2023

Engineering Science Research Opportunities Program (ESROP) - Global – University of Toronto

- Competitive fellowship supporting an international summer research placement.

01/05/2023

FrontierLab Mini – Osaka University

- Award supporting a short-term international research placement.

11/10/2019

Dean's Merit Award – University of Toronto

- Awarded for academic excellence, recognizing high-achieving incoming engineering students with outstanding admission averages.

PROJECTS

29/07/2024 – 25/04/2025
Open Source Tendon-Driven Continuum Robot (TDCR)

- Designed and fabricated an extensible TDCR with magnetically self-distributing spacer disks.
- Implemented real-time teleoperation for combined bending and extension.
- Developed experimental protocols for workspace, curvature, and follow-the-leader performance evaluation.

Link https://github.com/rozmarco/extensible_tdcr

30/11/2024 – 27/04/2025
AeroPick: Drone-Based Fruit Harvesting (ROB498 Capstone)

- Led the fruit detection subsystem for a harvesting drone designed with a small farm in Costa Rica, as part of a four-person capstone team.
- Fine-tuned a YOLO11 Nano detector with staged transfer learning (limes, then green fruit in foliage, then hand-labeled farm images), reaching 0.90 precision, 0.77 recall, and 0.83 F1 on held-out farm images.
- Deployed real-time detection on a Jetson Nano and integrated it with a servo-driven scissor-arm cutter, triggering cuts on confident, centered detections in the final demo.

Link https://github.com/CathyF9600/capstone_ws

CONFERENCES AND SEMINARS

13/05/2025 – 17/05/2025 Atlanta, United States
International Conference on Robotics and Automation (ICRA)

- Presented co-authored, published work during oral and poster sessions.

02/08/2025 – 07/08/2025 Montréal, Canada
Montréal Robotics Summer School with Professor Glen Berseth

- Selected for fully funded intensive program in manipulation, motion planning, RL, and perception.
- Team achieved top forward-walking performance in quadruped locomotion challenge.
- Hosted by Mila - Quebec Artificial Intelligence Institute.

02/2024 Oita, Japan
International Conference on Artificial Life and Robotics (ICAROB2024)

SKILLS

Continuum robot modeling | Optimization (continuous, discrete, nonlinear, mixed-integer) | Motion planning | Reinforcement learning (RL) | ROS/ROS2 | MuJoCo | Gazebo | Python (NumPy, SciPy, PyTorch) | MATLAB | C | CAD (Fusion 360) | Multi-material 3D printing | Git | Linux

VOLUNTEERING

30/09/2025
Robotic Competition Judge | FIRST Tech Challenge

- Evaluated robotics teams on technical design, system integration, and communication.
- Participated in award deliberations and technical interviews.

28/11/2019 – 29/01/2021
Engineering Mentor | Engineers Without Borders, University of Toronto Chapter

- Mentored high-school teams in engineering design competitions and community-impact programs.
- Guided students through problem formulation, design trade-offs, and prototype refinement.

INDUSTRY EXPERIENCE

26/04/2021 – 27/08/2022

Junior Software Developer | Linux Professional Institute

- Developed backend automation tools using PHP, Laravel, and Bash to support digital credential and exam generation workflows.
- Built utilities for automated image and JSON file creation and developed Grafana dashboards using SQL-based analytics.
- Updated internal codebases and documentation to support technical teams across deployment and reporting processes.

31/08/2021 – 14/07/2022

Robotics Technician | Brand M3dia Group

- Operated and programmed the company's AI robots, doubling user engagement over two months.
- Produced, choreographed, and filmed skits used for company marketing and custom event entertainment.
- Wrote current robot operating manuals and training tools now used as the company’s standard.

04/05/2020 – 28/08/2020

Software Developer Intern | Linux Professional Institute

- Implemented API endpoints and backend features using PHP and the Laravel framework.
- Developed tools for generating exam materials in PDF and XML formats.
- Collaborated with technical staff to update internal codebases and enhance reliability of exam delivery workflows.